

CoalSafe-MMTS: Multimodal Time-Series Machine Learning

Sub-Surface Coal Spontaneous Combustion Early Warning & Conflict Gating System

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GitHub Repository: <https://github.com/ManasPurTTY/CoalSafe-Sim-v2>

1. Executive Summary & Problem Overview

Spontaneous combustion in industrial coal stockpiles is a major hazard that begins deep sub-surface when trapped oxygen initiates exothermic low-temperature oxidation. Conventional monitoring systems suffer from two major limitations:

- **Sub-Surface Thermal Delay Gap:** By the time a surface probe or regular camera detects heat, the internal core has already reached critical thermal runaway ($T > 80^{\circ}\text{C}$).
- **Single-Sensor False Alarms:** A short-circuited gas probe or surface sunlight glare can trigger expensive, unnecessary automated water spray mitigation.

The Solution: We implemented **CoalSafe-MMTS**, an end-to-end Multimodal Deep Learning Model that combines **1-hour time-series sensor sequences** (gas and temperature dynamics) with **50×50 spatial thermal camera images**, coupled with an intelligent **Cross-Modal Conflict Detection Gate** that validates consensus (True) and suppresses false alarms (False).

2. 24-Hour Physics Simulation Expansion

The synthetic physical dataset was expanded from a 2-hour scale to a full **24-Hour (1,440-minute)** operational timeline across 8 calibrated scenarios (11,528 total rows):

- **Arrhenius Chemical Kinetics:** Scaled base reaction rate (0.00075 min^{-1}) with delta clipping $\Delta T \in [-1.0, 0.45]^{\circ}\text{C}$ guaranteeing 24-hour numerical thermodynamic stability.
- **Diurnal Day/Night Ambient Cycle:** Realistic continuous sinusoidal temperature swing ($\pm 6^{\circ}\text{C}$ swing, with minimum at 06:00 AM and peak heat at 14:00 / 02:00 PM).
- **10-Minute Periodic Image Sampling:** Generates 145 thermal frames per scenario ($t = 0, 10, 20, \dots, 1440$) and corresponding snapshot graphs under `graphs/per_minute/`.

3. Multimodal ML Model Architecture (`CoalSafe-MMTS`)

The neural network contains 4 integrated processing blocks designed for time-series and spatial perception:

1. Tabular Time-Series Branch (Sensor Reader)

Takes a sliding history window of 6 consecutive time steps (60 minutes) across 10 physical sensor channels: ambient_temp, humidity, wind_speed, coal_moisture, CO, CO2, O2, surf_temp_mean, surf_temp_max, heating_rate.

- **2-Layer Bi-directional GRU (hidden size = 64):** Reads sequences forward and backward to detect rates of gas accumulation.
- **Temporal Self-Attention:** Automatically calculates importance weights across the 60-minute window, highlighting moments of rapid gas acceleration.
- **Output:** 128-dimensional latent vector (h_{tab}).

2. Spatial Thermal Image Branch (Camera Reader)

Takes the 50×50 pixel 3-channel thermal heatmap image corresponding to the current time step.

- **3-Stage 2D-CNN:** Conv2D layers ($32 \rightarrow 64 \rightarrow 128$ filters) with LeakyReLU activations and MaxPool downsampling.
- **Adaptive Average Pooling:** Reduces spatial feature maps into a dense 128-dimensional latent vector (h_{img}).

3. Cross-Modal Gated Bilinear Fusion (The AI Brain)

Fuses the 128 sensor features and 128 thermal camera features into a unified 256-dimensional representation:

- **Adaptive Gating Score ($g \in [0, 1]$):** $g = \sigma(W_g [h_{\text{tab}}; h_{\text{img}}])$ dynamically weighs sensor vs. thermal reliability.
- **Interaction Synthesis:** Fuses $[g \odot h_{\text{tab}}; (1-g) \odot h_{\text{img}}; h_{\text{tab}} \odot h_{\text{img}}]$ via an MLP with Dropout (0.2).

4. Multi-Head Outputs & Conflict Gating Module

- `tab_head`: Auxiliary prediction from sensors only ($\hat{y}_{\text{tab}} \in [0, 100\%]$).
- `img_head`: Auxiliary prediction from thermal images only ($\hat{y}_{\text{img}} \in [0, 100\%]$).
- `fused_reg_head`: Unified Multimodal Risk Score ($\hat{y}_{\text{fused}} \in [0, 100\%]$).
- `fused_class_head`: 5-Class Categorical Classifier (NORMAL, LOW, MEDIUM, HIGH, CRITICAL).

4. The "True vs. False" Conflict Gating Benchmarks

As requested by the project requirements, the system computes the **Modality Disagreement**: $\Delta_{\text{modal}} = |\hat{y}_{\text{tab}} - \hat{y}_{\text{img}}|$. If $\Delta_{\text{modal}} \leq 25\%$, the decision is verified as **TRUE (Valid Consensus)**. If $\Delta_{\text{modal}} > 25\%$, the decision is flagged as **FALSE (Conflict Detected)** to suppress false alarms.

Test Scenario	Input Conditions	Sensor Risk	Thermal Risk	Disagreement	Consensus Status	Action Taken
Case 1: Normal Baseline	Normal gases (CO < 5 ppm) + Cold camera image	10.3%	7.3%	3.0%	TRUE (Valid)	Routine Monitoring
Case 2: Runaway Fire	High CO gas (85 ppm) + Hot thermal hotspot image	66.4%	66.9%	0.5%	TRUE (Valid)	Automated Water Spray Triggered
Case 3: Injected Sensor Glitch	Glitch spikes CO to 250 ppm, but Thermal Camera is Cold	67.4%	7.3%	60.0%	FALSE (Conflict)	Spray Suppressed (Sensor Fault Flag)
Case 4: Camera Glare Injected	Normal gases, but sunlight glare on Thermal Camera	10.3%	66.9%	56.6%	FALSE (Conflict)	Spray Suppressed (Optical Glare Flag)

5. Codebase Deep Dive (File-by-File Summary)

1. code/ml/dataset.py (Data Loader & Sliding Window)

Loads `sensor_data.csv` and `thermal_metadata.csv`, handles missing values using `ffill()` and `bfill()`, calculates Z-score feature normalization $(x - \mu) / \sigma$, generates 60-minute sliding sequence windows, and formats data into PyTorch batches of size 16.

2. `code/ml/model.py` (MultimodalTimeSeriesNet)

Defines the complete neural network architecture: 2-layer Bi-GRU with Temporal Attention, 3-layer Conv2D spatial extractor, Gated Bilinear Fusion, multiple prediction heads, and the `evaluate_consensus()` gating function.

3. `code/ml/train.py` (GPU Training Pipeline)

Trains the 520,713 model parameters over 40 epochs using AdamW optimizer ($\text{lr} = 10^{-3}$, weight decay $= 10^{-4}$) and Cosine Annealing learning rate schedule. Uses a multi-task loss combining Smooth L1 regression loss, Cross-Entropy classification loss, and modality consistency regularization. Saves the best checkpoint to `coalsafe_mmts_best.pt`.

4. `code/ml/evaluate.py` (Testing & Conflict Benchmarks)

Evaluates the trained model on unseen Scenario S07 (Late Mitigation) and executes the 4 benchmark conflict experiments demonstrating 100% false-alarm suppression.

5. `code/ml/plot_ml_results.py` (Visualization Generator)

Generates presentation-grade figures under `graphs/ml_results/`: training convergence curves, 24-hour S07 test prediction trajectories, and conflict gating bar charts.

6. Key Experimental Results

- **Training Hardware:** NVIDIA GeForce GTX 1650 GPU (CUDA acceleration)
- **Total Trainable Parameters:** 520,713
- **Best Validation MAE: 3.33%** (Validation Accuracy: **88.3%**)
- **Held-out Test Scenario S07:** $R^2 = \mathbf{0.8321}$, $\text{MAE} = 6.57\%$, $\text{RMSE} = 8.49\%$
- **False Alarm Protection:** 100% success rate on detecting and suppressing injected sensor faults and camera glare.

7. Viva & Teacher Question-Answer Cheatsheet

Q1: Why did you use a Multimodal model instead of only temperature sensors?

Answer: Sub-surface coal fires start deep underground. Surface temperature remains cold for hours due to coal's thermal insulation. Gas sensors (CO) detect early oxidation, while thermal cameras provide exact spatial coordinates. Combining both gives early warning with spatial precision.

Q2: Why did you choose GRU over standard Feedforward networks for sensors?

Answer: Sensor data is time-series. A single static CO reading of 15 ppm is ambiguous, but observing CO accelerate from 2 ppm to 15 ppm over 1 hour indicates active spontaneous combustion. GRU captures temporal rates of rise.

Q3: How does the Conflict Gate protect the system?

Answer: The model computes individual risk estimates for sensors ($\hat{y}_{\text{tab}}$) and camera ($\hat{y}_{\text{img}}$). If $|\hat{y}_{\text{tab}} - \hat{y}_{\text{img}}| > 25\%$, the model outputs `False` (Conflict), preventing false automated water spray activation during sensor glitches or solar camera glare.